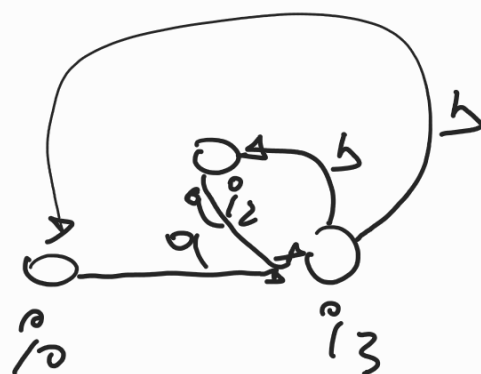
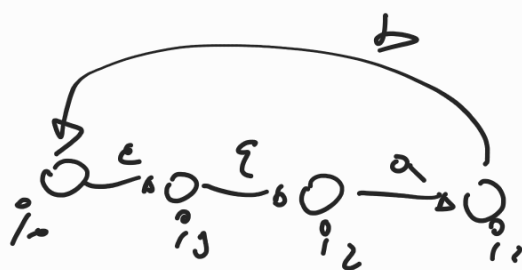
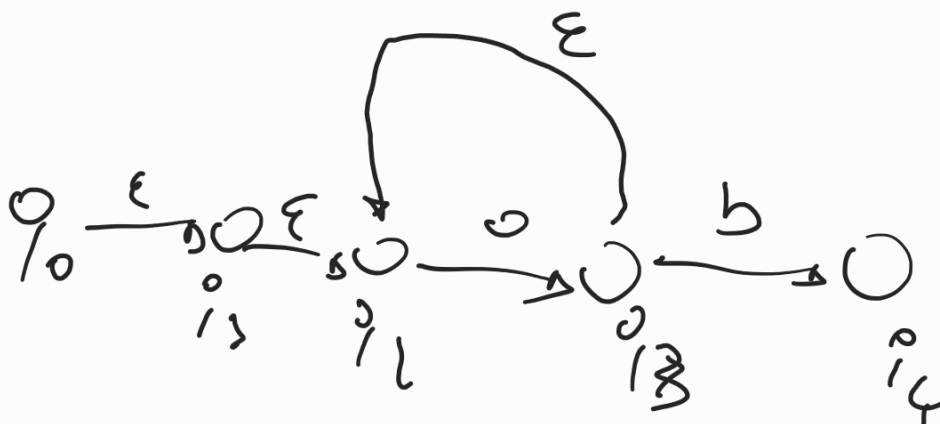
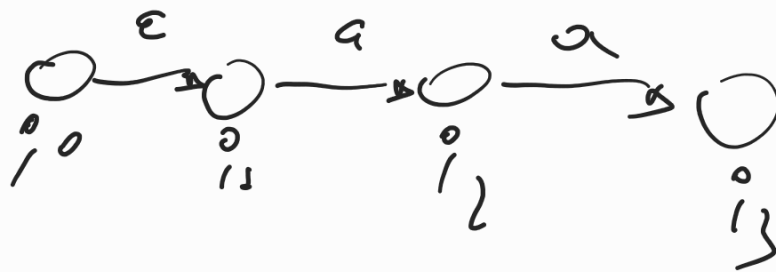
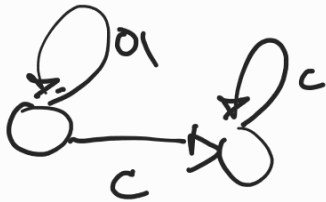


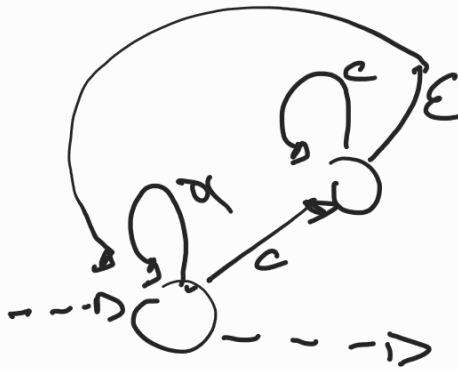


$$r = ((a^*c^+)^*b \cup ac^+b)^+$$

$$a^*c^+$$

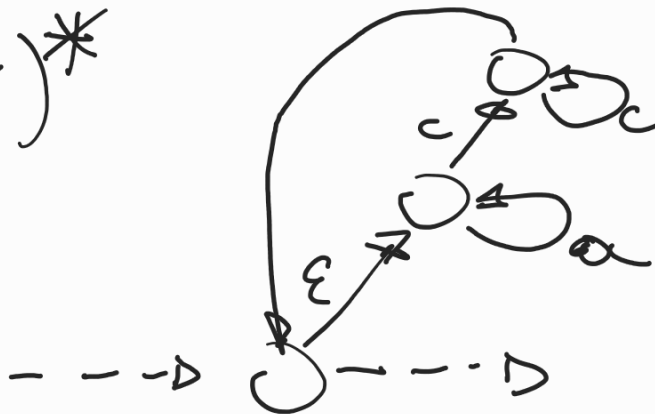


$$(a^*c^+)^*$$



$$a^* \cup (a^*c^+)^*$$

$$(a^*c^+)^*$$

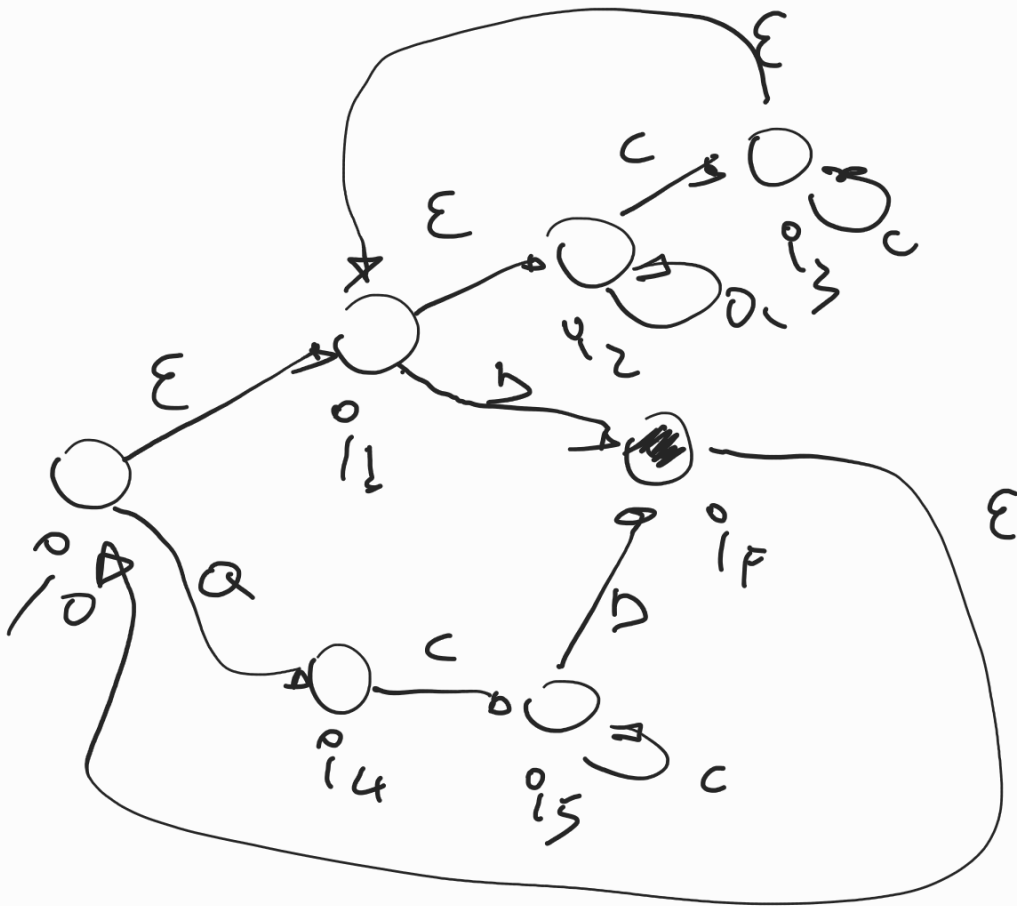


$$(0^*c^*)^*$$

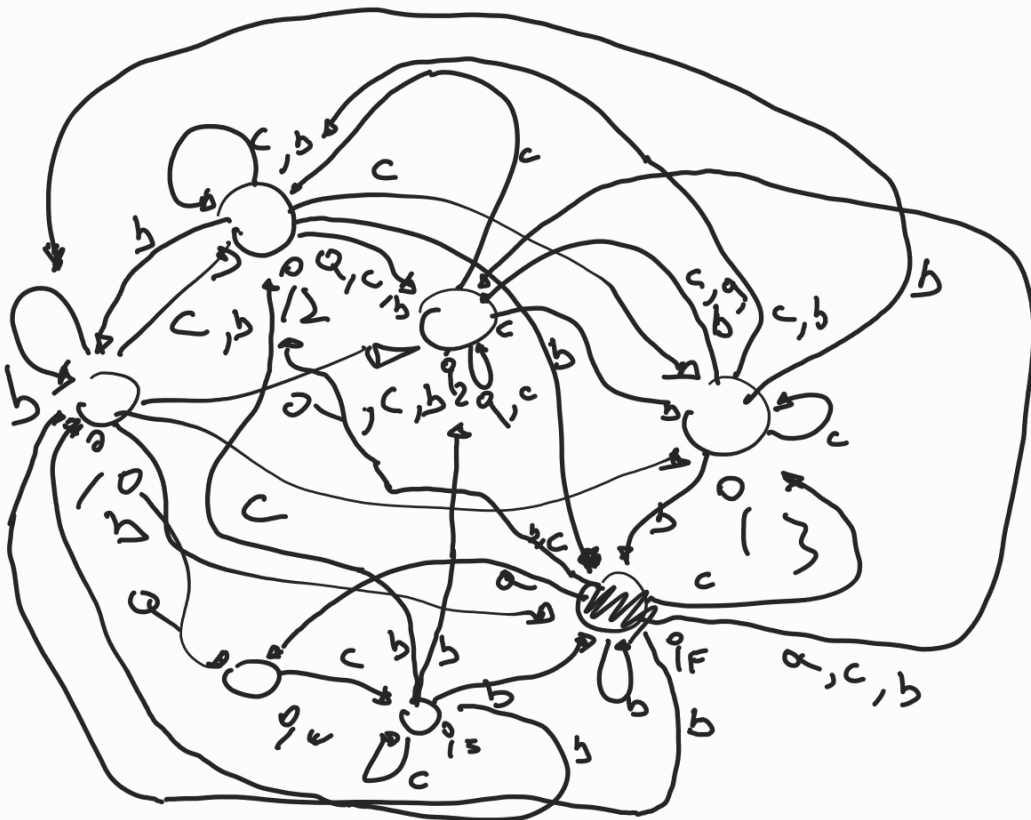


$$(0^*c)^*$$

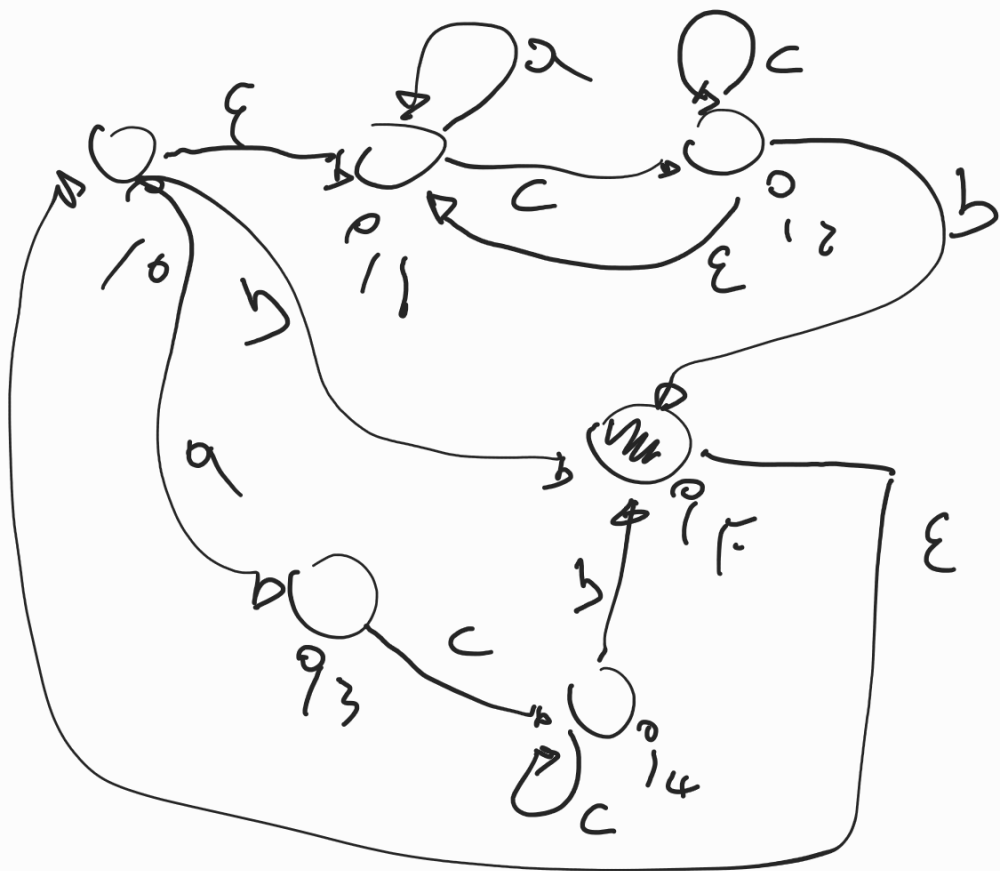
$$r = ((a^* - c^+)^* b \cup a c^+ b)^+$$



ELIM. ϵ -TRANS



$$r = ((a^*c^+)^*b \cup ac^+b)^+$$



$S_0 \rightarrow S$
 $S \rightarrow T$
 $T \rightarrow a T a$
 $T \rightarrow b T b$
 $T \rightarrow c L$
 $L \rightarrow \epsilon$
 $L \rightarrow c L$

$S_0 \rightarrow S$	$\{ \}$
$S \rightarrow \cdot T$	$\{ \}$
$T \rightarrow a \cdot T a$	$\{ \}$
$T \rightarrow b \cdot T b$	$\{ \}$
$T \rightarrow c \cdot L$	$\{ \}$

$S_0 \rightarrow S \cdot \{ \}$

$\bar{I}_2$

$S_0 \rightarrow S \cdot \{ \}$

$\bar{I}_3$

T

$S \rightarrow T \cdot \{ \}$

$\bar{I}_4$

a

$T \rightarrow a \cdot T a$	$\{ \}$
$T \rightarrow a \cdot a T a$	$\{ a \}$
$T \rightarrow a \cdot b T b$	$\{ a \}$
$T \rightarrow a \cdot c L$	$\{ a \}$

$\bar{I}_5$

$T \rightarrow c \cdot L$	$\{ \}$
$L \rightarrow \cdot$	$\{ \}$
$L \rightarrow c \cdot L$	$\{ \}$

$\bar{I}_7$

T

a

b

c

$\bar{I}_{11}$

$\bar{I}_{13}$

$\bar{I}_{14}$

$\bar{I}_{15}$

L

$T \rightarrow c L \cdot \{ \}$

$\bar{I}_8$

$L \rightarrow c \cdot L$	$\{ \}$
$L \rightarrow \cdot$	$\{ \}$
$L \rightarrow c \cdot L$	$\{ \}$

L

$L \rightarrow c L \cdot \{ \}$

$\bar{I}_{16}$

$T \rightarrow b \cdot T b$	$\{ \}$
$T \rightarrow a \cdot a T a$	$\{ b \}$
$T \rightarrow a \cdot b T b$	$\{ b \}$
$T \rightarrow a \cdot c L$	$\{ b \}$

a b c

$T \rightarrow b T \cdot b \{ \}$

b

$T \rightarrow b T b \cdot \{ \}$

$\bar{I}_{20}$

$\boxed{\bar{I}_5} \xrightarrow{T}$

$\boxed{T \rightarrow a \mid T \cdot a \mid \{ \}} \quad \bar{I}_{11}$

$\xrightarrow{a}$

$\boxed{T \rightarrow a \mid T \cdot a \cdot \{ \}} \quad \bar{I}_{12}$

$\boxed{\bar{I}_5} \xrightarrow{a} \boxed{\bar{I}_{23}} \xrightarrow{a}$

$\boxed{T \rightarrow a \mid T \cdot a \mid \{ a \}} \quad \bar{I}_{13}$
 $T \rightarrow a \mid T \cdot a \mid \{ a \}$
 $T \rightarrow a \mid T \cdot b \mid \{ a \}$
 $T \rightarrow a \mid T \cdot c \mid \{ a \}$

$\xrightarrow{T}$

$\boxed{T \rightarrow a \mid T \cdot a \mid \{ a \}} \quad \bar{I}_{14}$

$\xrightarrow{a}$

$\boxed{T \rightarrow a \mid T \cdot a \mid \{ a \}} \quad \bar{I}_{15}$

$\boxed{\bar{I}_5} \xrightarrow{b} \boxed{\bar{I}_{23}} \xrightarrow{b}$

$\boxed{T \rightarrow b \mid T \cdot b \mid \{ a \}} \quad \bar{I}_{16}$
 $T \rightarrow b \mid T \cdot a \mid \{ b \}$
 $T \rightarrow b \mid T \cdot b \mid \{ b \}$
 $T \rightarrow a \mid T \cdot c \mid \{ b \}$

$\xrightarrow{a} \boxed{\bar{I}_{23}} \xrightarrow{b} \boxed{\bar{I}_{24}} \xrightarrow{c} \boxed{\bar{I}_{25}}$

$\boxed{T \rightarrow b \mid T \cdot b \mid \{ a \}} \quad \bar{I}_{17}$

$\boxed{T \rightarrow b \mid T \cdot b \mid \{ a \}} \quad \bar{I}_{18}$

$\boxed{\bar{I}_5} \xrightarrow{c} \boxed{\bar{I}_{22}} \xrightarrow{c}$

$\boxed{T \rightarrow c \mid T \cdot c \mid \{ a \}} \quad \bar{I}_{19}$
 $T \rightarrow c \mid T \cdot a \mid \{ a \}$
 $T \rightarrow c \mid T \cdot c \mid \{ a \}$

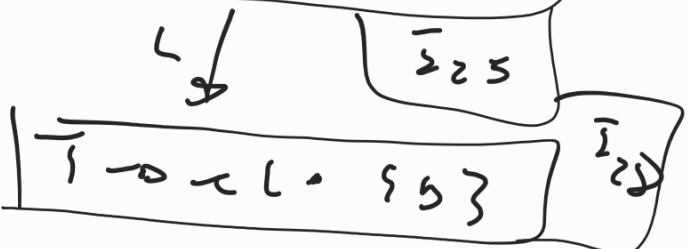
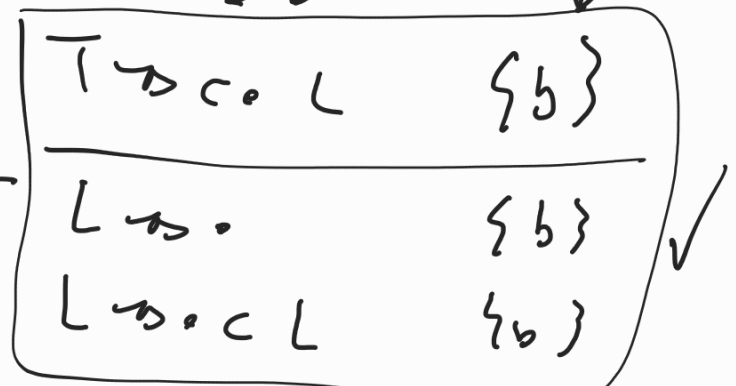
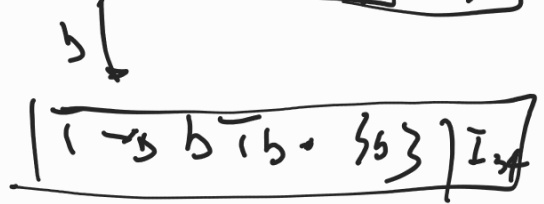
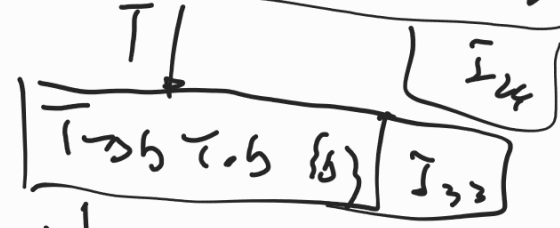
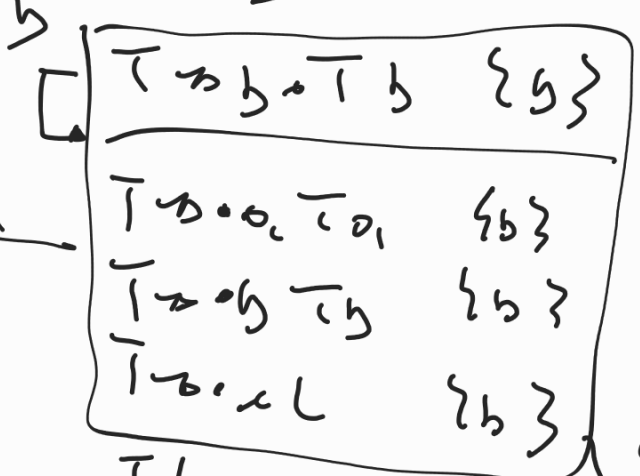
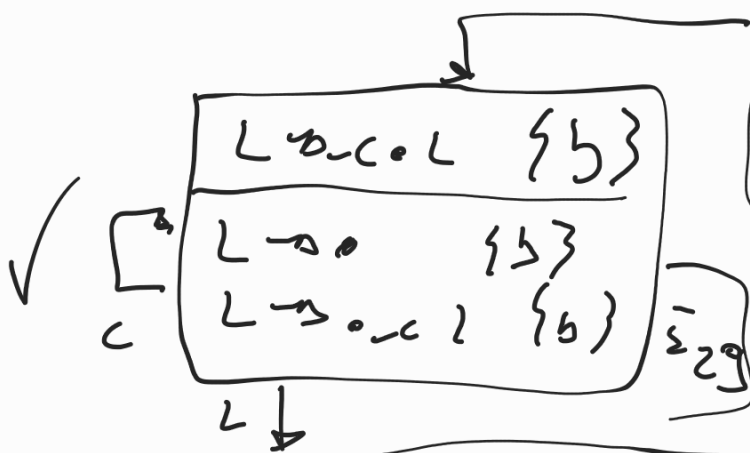
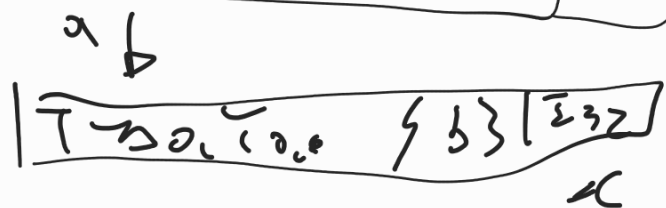
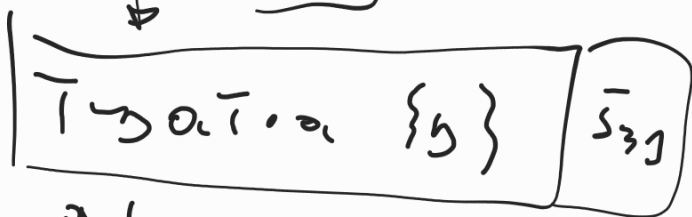
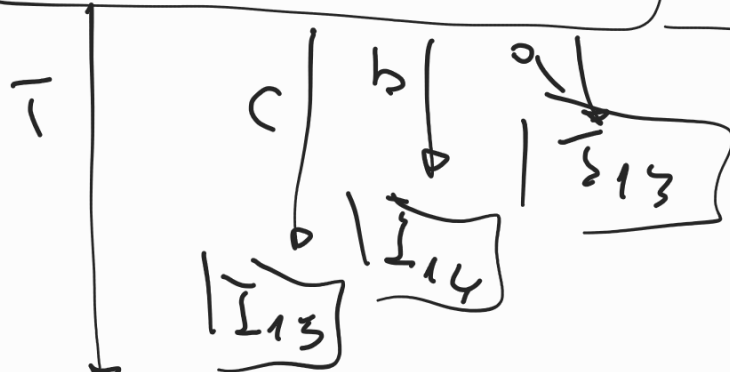
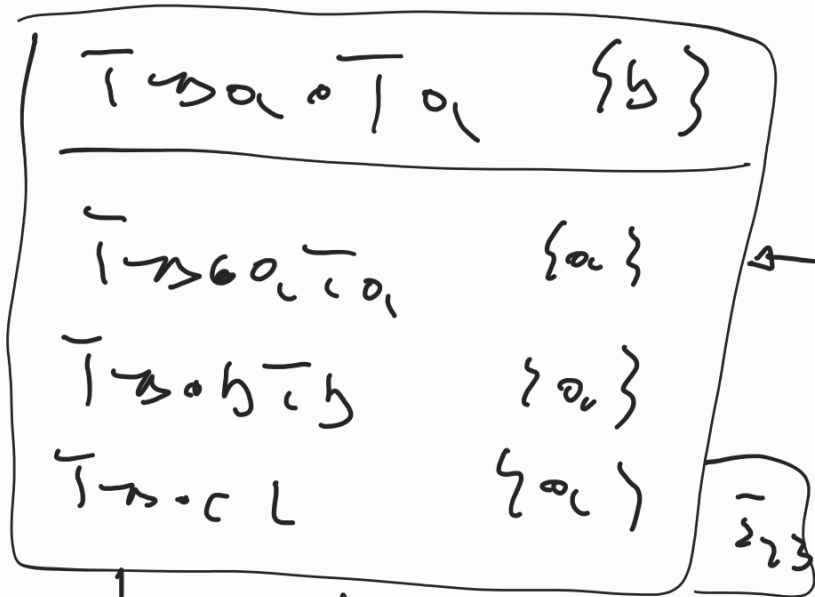
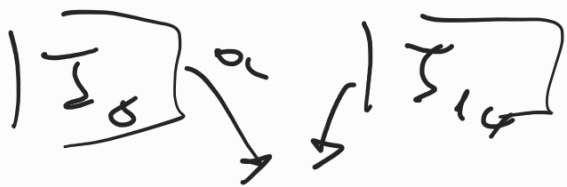
$\boxed{T \rightarrow c \mid T \cdot c \mid \{ a \}} \quad \bar{I}_{20}$

$\boxed{T \rightarrow c \mid T \cdot c \mid \{ a \}} \quad \bar{I}_{21}$
 $T \rightarrow c \mid T \cdot a \mid \{ a \}$
 $T \rightarrow c \mid T \cdot c \mid \{ a \}$

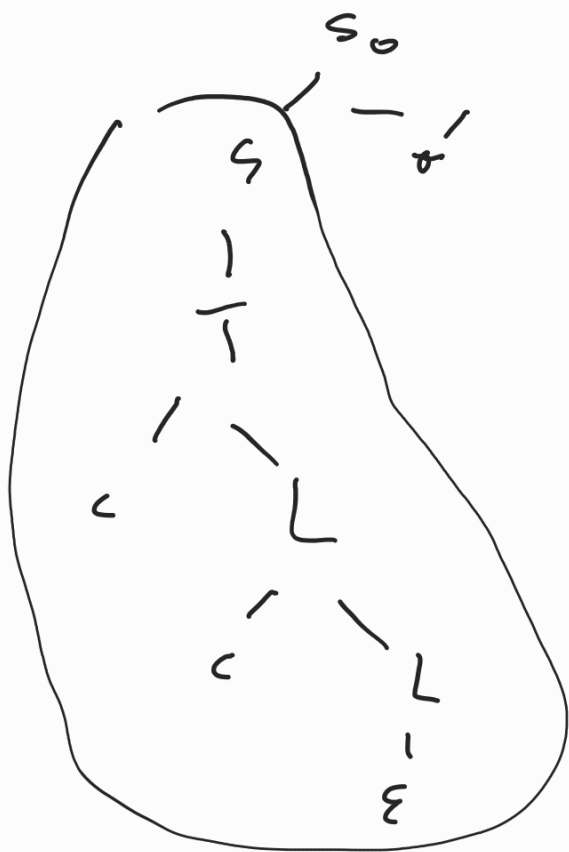
$\boxed{T \rightarrow c \mid T \cdot c \mid \{ a \}} \quad \bar{I}_{22}$

$\xrightarrow{c}$

$\boxed{T \rightarrow c \mid T \cdot c \mid \{ a \}} \quad \bar{I}_{23}$



$c \in LR(I)$



$S_0 \rightarrow S \checkmark$

$S \rightarrow RS$

$S \rightarrow \epsilon$

$R \rightarrow L d v$

$C \rightarrow a L$

$C \rightarrow v$

$L \rightarrow C X$

$X \rightarrow S C X$

$X \rightarrow \epsilon$

$S_0 \rightarrow S \checkmark$	$\emptyset$
$S \rightarrow RS$	$\{ \checkmark \}$
$S \rightarrow \epsilon$	$\{ \checkmark \}$
$R \rightarrow L d v$	$\{ \checkmark, a, v \}$
$L \rightarrow C X$	$\{ \checkmark \}$
$C \rightarrow a L$	$\{ a, v \}$
$C \rightarrow v$	$\{ a, v, d \}$

$\bar{S}_1$